A. Wikipedia-derived Local Graph

Expand Wikipedia links into a local rooted knowledge graph

Law Games Dining Travel
Finance Education Design Health
24 domains ...

WIKIPEDIA
Root entity

Wikipedia
Whale shark
Root
The whale shark is a slow-moving, filter-feeding carpet shark. The species was distinguished in April 1828 after the harpooning of a specimen in Table Bay.

Wikipedia
Filter-feeding carpet shark

Wikipedia
Table Bay

Subgraph sampling

S1
S2
S3

B1. Query Construction

S_i

LLM synthesis

Query q

B2. Rubrics Construction

Generate a rubric for each edge in the sampled subgraph

S_i

Each sampled edge e with linked evidence pages
Source page
Target page

LLM synthesis

Rubric 1
Rubric

All edges in S

Rubric 5
Rubric 4
Rubric 3
Rubric 2
Rubric 1
Rubric set R_s

C. Topology-aware Coverage Evaluation

Judge coverage on the sampled subgraph with structural metrics

Query q
+
Deep Research Agents
Report
Judge LLM
Per-edge coverage judgment
Covered
Uncovered
Scored rubric graph
Breadth B
Depth D
Bottom-up Recursive Aggregation
Overall Coverage
$S = \sqrt{B \cdot D}$